A Cross-Scene Self-Representative Network for Hyperspectral Band Selection

Weiwei Sun, *Senior Member, IEEE*, Ke He, Gang Yang, *Member, IEEE*, Jiangtao Peng, *Senior Member, IEEE*, Kai Ren, *Student Member, IEEE*, and Jiancheng Li

***Abstract*— This article proposes a novel deep learning-based framework for hyperspectral band selection, named Cross-Scene Self-Representative Network (CSSRnet). The proposed method leverages the rich labels of the source domain (SD) to guide the band selection in the target domain (TD). To the best of our knowledge, CSSRnet is the first deep-learning-based solution for cross-scene hyperspectral band selection. First, the CSSRnet employs contextual attention mechanism to capture the latent features of SD and TD. It combines the self-attention mechanism with convolutional operations to capture static and dynamic contextual information. Then, the self-representative layer provides the self-representative coefficient of SD and TD. Subsequently, the maximum mean difference (MMD) is utilized to align the self-representative coefficients of both SD and TD. To enhance the representativeness and precision of these coefficients, we introduce different tasks for the SD and TD branches. Finally, a suitable band subset is selected based on a ranking method that evaluates each band's importance by considering its self-representative coefficient matrix. Experiments are carried out to assess the efficacy of CSSRnet. These experiments focus on evaluating classification accuracy across various cross-scene datasets, the utility of cross-scene concepts, and the practical application in coastal wetland. Experimental results confirm the effectiveness of CSSRnet.**

***Index Terms*— Band selection, cross-scene, hyperspectral image (HSI), self-representative.**

I. Introduction

As a significant advancement in earth observation, hyperspectral image (HSI) captures continuous and narrow bands with high spectral resolution [1]. The HSI is capable of simultaneous acquisition of 2-D spatial and 1-D spectral information from the target, providing a comprehensive carrier for abundant spatial and spectral information [2], [3]. Therefore, HSI has found extensive applications in environmental monitoring [4], coastal wetland mapping [5], resource exploration [6], and so on. However, hyperspectral data pose challenges and inefficiencies in processing due to their high dimensionality, information redundancy, and limited labeled samples [7], [8], [9]. These issues result in high time complexity, overfitting phenomenon, and Hughes phenomenon.

The method of dimensionality reduction can effectively reduce data dimensionality, thereby enhancing processing efficiency and subsequent applications [10]. In contrast to feature extraction, band selection involves selecting a subset of bands from the original set preserving the spectral meaning [11]. Currently, band selection methods can be categorized into unsupervised and supervised approaches [12], [13]. The unsupervised band selection method relies primarily on the inherent characteristics of HSI. It employs techniques, such as clustering-based [14] and sparsity-based [15] to explore structure of HSI, ultimately selecting the most representative bands. These approaches exhibit high universality and flexibility as it obviates the need for labeled samples. However, the classification accuracy of unsupervised band selection method is often inferior to that of supervised band selection method due to the absence of labeled samples [16], [17]. The supervised band selection method aims to identify the most representative bands by leveraging information from labeled samples [18], [19]. It can be employed to identify more distinctive spectral bands. However, the labeled samples in HSI necessitate a time-consuming and labor-intensive acquisition process, thereby limiting its practical applicability [20]. The conflict between the classification accuracy of the selected bands and the limited labeled samples poses a challenge to current band selection methods.

Cross-scene learning is a transfer learning method. In the field of remote sensing, the core idea of cross-scene learning is to transfer knowledge from the source domain (SD) to the target domain (TD) [21], [22], [23]. The information learned in SD can be regarded as a kind of prior knowledge, facilitating faster convergence and enhancing the performance of the model in TD [24], [25].

Using cross-scene learning for hyperspectral band selection can effectively alleviate the dilemma of single-scene band selection. In practice, HSI may only have a very small number of labeled samples available. In another scene, there may be

Manuscript received 6 October 2023; revised 21 January 2024; accepted 29 January 2024. Date of publication 6 February 2024; date of current version 23 February 2024. This work was supported in part by Youth Scientist Project National Key Research and Development Program of China under Grant 2023YFF1305600, in part by the National Natural Science Foundation of China under Grant 42122009, in part by Zhejiang Province "Pioneering Soldier" and "Leading Goose" Research and Development Project under Grant 2023C01027, in part by the Public Projects of Ningbo City under Grant 2021S089, and in part by Ningbo Science and Technology Innovation 2025 Major Special Project under Grant 2021Z107 and Grant 2022Z032. *(Corresponding author: Gang Yang.)*

Weiwei Sun, Ke He, and Gang Yang are with the Department of Geography and Spatial Information Techniques, Ningbo University, Ningbo 315211, China (e-mail: sunweiwei@nbu.edu.cn; hk2098@yeah.net; yanggang@nbu.edu.cn).

Jiangtao Peng is with the Hubei Key Laboratory of Applied Mathematics, Faculty of Mathematics and Statistics, Hubei University, Wuhan 430062, China (e-mail: pengjt1982@hubu.edu.cn).

Kai Ren is with the Faculty of Electrical Engineering and Computer Science, Ningbo University, Ningbo 315211, China (e-mail: kairen0908@outlook.com).

Jiancheng Li is with the School of Geodesy and Geomatics, Wuhan University, Wuhan 430062, China (e-mail: jcli@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3362994

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:41:19 UTC from IEEE Xplore. Restrictions apply.

a sufficient number of labeled samples with similar type of ground object to the current scene. The shared information between these scenes can be used to assist labeled samples in the current scene for band selection. The single-scene band selection methods focus solely on a single scene and neglect the information in similar scenes. The information from multiple HSI scenes can be simultaneously utilized to select optimal band subsets [26]. However, due to the influence of atmospheric conditions and imaging environment, the spectral distribution may change, resulting in spectral drift [27]. Simultaneously, cross-scene learning necessitates careful consideration of the disparities between SD and TD. The techniques of feature extraction, model training, and distribution alignment should be adapted to suit the new scene.

To mitigate the aforementioned issues, researchers have made some attempts. cross-domain information gain (CDIG) [28] was proposed to consider the consistency of features in different scenes and the ability of features to distinguish between different types of features. Cross-domain ReliefF (CDRF) [29] extended the traditional ReliefF method into a cross-domain feature selection method for HSI. CDRF used the information of SD and TD to sort the weight of each feature and select the optimal band subset. Based on CDRF, I-ReliefF was extended to a cross-scene band selection algorithm, namely, cross-domain I-ReliefF (CDIRF) [30]. Cross-Domain Hybrid Whale Optimization Algorithm with Simulated Annealing algorithms (CDWOASA) [31] was proposed. It carried out simulated annealing hybrid whale optimization algorithm for SD and TD, respectively, and used the information of two HSI scenes to select discriminant and scene invariant feature subsets. However, these methods solely examined the shallow layer characteristics and failed to delve into the deep characterization of HSI. The internal structure of HSI is intricate, and relying solely on shallow layer information may result in imprecise selection of domain-invariant bands, which hinders subsequent utilization.

To solve the above problems, this article proposes a Cross-Scene Self-Representative Network (CSSRnet) for hyperspectral band selection. CSSRnet selects bands based on the end-to-end deep learning framework. It employs a similar HSI scene to guide band selection of TD. This method introduces the idea of domain adaptation to reduce the data distribution difference and feature deviation between SD and TD.

In order to intensify feature representation, CSSRnet integrates a novel contextual attention technique to identify potential features of SD and TD. This ingenious contextual attention mechanism blends the self-attention mechanism with convolutional operations to efficiently capture static and dynamic contextual information. Subsequently, the self-representative layer is utilized to generate the self-representative coefficients of SD and TD. To alleviate the feature disparity between SD and TD, we implement a distribution alignment loss function to reduce the potential divergence in their representations. Maximum mean difference (MMD) methods are adopted to align the self-representative coefficients of SD and TD. To foster more representative coefficients, auxiliary losses are incorporated into the SD and TD branches. A classification loss is included in the SD branch to refine the recognition capability of extracted features for ground objects. Simultaneously, the entropy minimization loss function is incorporated in the TD branch to judiciously measure the class overlap of the target samples. Finally, a suitable band subset is chosen based on the self-representative coefficient matrix using the sorting method. Our main contributions are as follows.

1) An end-to-end deep learning framework for cross-scene hyperspectral band selection is proposed in this article. The framework introduces the concept of domain adaptation and leverages the rich label information in SD to guide the band selection for TD. Meanwhile, CSSRnet employs deep learning to address the limitation of existing methods that only focus on shallow features of HSI and instead delves into its deeper characteristics.
2) CSSRnet proposes a novel attention mechanism. It integrates the neighbor information of the key with its own features to generate attention maps. Moreover, the self-representative layer is used to produce self-representative coefficients. Additionally, auxiliary losses are incorporated to learn these coefficients more accurately.

The following article is structured as follows. Section II presents the CSSRnet approach, followed by a series of experiments on various datasets in Section III. Section IV provides concluding remarks of this article.

II. Proposed Method

Fig. 1 illustrates the proposed CSSRnet for HSI band selection. HSI is a 3-D data cube with $h \times w$ pixels and b bands, and data from SD and TD can be represented by the following formula: $X_s = \{\mathbf{x}_i^s\}_{i=1}^{h_s \times w_s} \in R^b$ and $X_t = \{\mathbf{x}_i^t\}_{i=1}^{h_t \times w_t} \in R^b$.

A. Contextual Attention Layer

The transformer-style attention mechanism has demonstrated remarkable performance in various natural language processing and computer vision tasks. Researchers are currently exploring the potential of attention mechanisms in hyperspectral applications, with various types of attention mechanisms having been successfully implemented across different hyperspectral fields, such as band selection and classification [32], [33]. However, previous designs have primarily relied on the similarity between query-key pairs in isolation to generate attention maps, while overlooking the rich contextual information among neighboring.

In order to enhance the representative power in transformer-style attention module, we design a novel contextual attention mechanism based on previous research [34] to extract potential representations from both SD and TD data. This novel attention mechanism integrates contextual information mining with self-attentional learning, obviating the need for additional branches to mine contextual information between keys. This contextual attention mechanism maps contextual key information and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:41:19 UTC from IEEE Xplore. Restrictions apply.

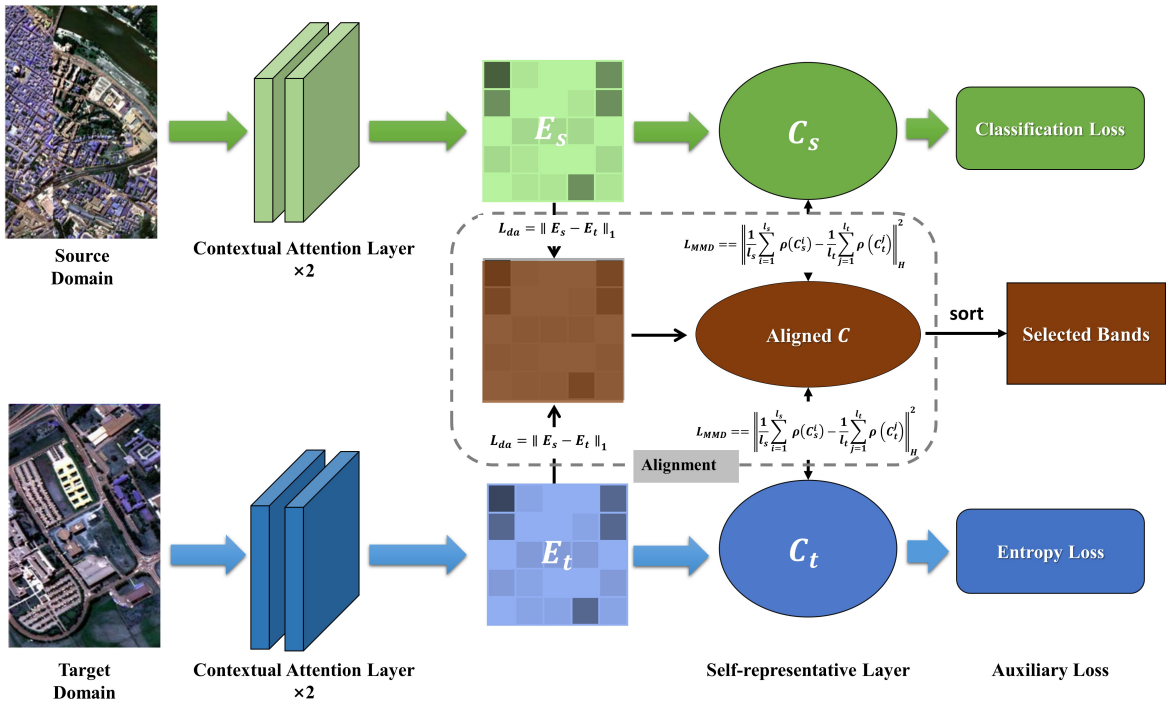


Fig. 1. Overall flowchart of CSSRnet for HSI band selection.

itself key information through 3 × 3 convolution operation and 1 × 1 convolution operation, forming more representative keys from different scales. Two 1 × 1 convolution operations can enhance the characterization ability of the bands. It can be viewed as the squeeze-and-excitation to form attention features within the bands.

The structure of the novel attention layer proposed in this article is shown in Fig. 2. The original HSI is sent to this new contextual attention layer, and k_s, q, and v required for subsequent operations are first obtained by the embedding function. k_s is extracted by a convolution operation with a convolution kernel size of 3 × 3 and a convolution operation with a convolution kernel size of 1 × 1, and v is obtained by an embedding function with a convolution kernel size of 1 × 1

$$k_s = X\theta_{k2} + X\theta_{k1} \tag{1}$$

$$q = X \tag{2}$$

$$v = X\theta_v \tag{3}$$

where θ_{k2}, θ_{k1}, and θ_v represent the embedded matrix of k_s and v, respectively; each convolution operation is followed by a batch norm (BN) and a rectified linear unit (ReLU). As k_s aggregates the information of the neighborhood, it can be called a static contextual feature.

In order to enhance the representative ability of the attention mechanism, the attention matrix is obtained through two convolution operations with the size of 1 × 1 convolution kernel

$$A = \text{att}(\text{cat}(k_s, q); \theta_a) \tag{4}$$

where att(·) represents two 1 × 1 convolution operations for learning the attention matrix, cat(·) is the stack operations for k_s and q, and θ_a is the parameter to be trained. This approach

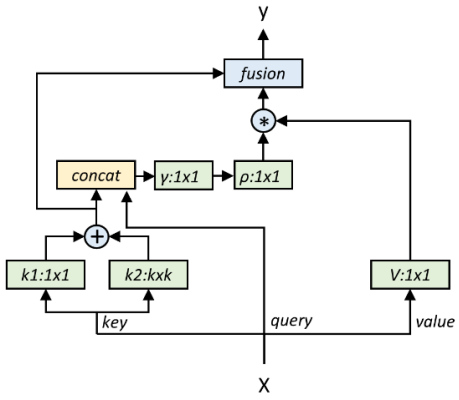


Fig. 2. Proposed new contextual attention layer.

takes into account the contextual information of the key and the information of itself rather than multiplying the query-key pairs in isolation at each location to calculate the attention matrix.

After that, the learned attention matrix A is used for dynamic aggregation of input features to realize the interaction of features. Specifically, the attention matrix A is used to multiply v to produce a dynamic context-aware representation of features. Finally, the dynamic context representation k_d is added to the static context representation k_s as the final output

$$k_d = A \circledast v \tag{5}$$

$$E = k_s + k_d. \tag{6}$$

B. Self-Representative Layer

The self-representative layer uses the inherent properties of the data itself to represent the input data. For a given dataset

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:41:19 UTC from IEEE Xplore. Restrictions apply.

$G = \{g_i\}_{i=1}^{m} \in R^{d\times m}$, we assume that it belongs to a union of n subspaces, namely, $\{S_i\}_{i=1}^{n}$, and the dimensions of n subspaces are $\{d_i\}_{i=1}^{n}$. Any sample g_i in the subspace can be expressed as a linear combination of other samples in the same subspace, that is, the "self-representative" property of data [35], [36], [37], which can be expressed as follows:

$$\text{Min } \|C\|_p, \quad \text{s.t. } G = GC, \quad \text{diag}(C) = 0 \tag{7}$$

where C is the self-representative matrix, $\|\cdot\|_p$ is an arbitrary matrix norm, and $\text{diag}(C) = 0$ constrains the diagonal element of the self-representative matrix to be equal to 0 to avoid trivial solutions [38].

CSSRnet uses the self-representative layer to learn the self-representative coefficient end to end. The features extracted from the new context attention layer are sent to the self-representative layer to obtain the self-representative coefficient of SD and TD. The self-representative coefficient can be obtained by optimizing the following loss function:

$$L_{\text{se}} = \frac{1}{2}\|E - EC\|_2^2 + \frac{\lambda}{2}\|C\|_2^2, \quad \text{s.t. } \text{diag}(C) = 0 \tag{8}$$

where $\|\cdot\|_2$ indicates the $L2$ norm, so as to avoid the high correlation of HSI leading to the "too sparse" coefficient solution, and λ is the tradeoff coefficient.

C. Domain Alignment

The contextual attention layer extracts the features in SD and TD as E_s and E_t, respectively. To achieve cross-scene transfer, features need to be aligned, so that SD and TD features have similar distributions. This process can be achieved by the following distribution alignment loss function:

$$L_{\text{da}} = \|E_s - E_t\|_1 \tag{9}$$

where $\|\cdot\|_1$ is the $L1$ norm. The purpose of this function is to align the attention features and to lay a foundation for the subsequent matching of self-representative coefficients.

With regard to minimize the difference between the self-representative coefficients of SD and TD, MMD is used to align the self-representative coefficients

$$L_{\text{MMD}} = \left\| \frac{1}{l_s}\sum_{i=1}^{l_s} \rho\left(C_s^i\right) - \frac{1}{l_t}\sum_{j=1}^{l_t} \rho\left(C_t^i\right) \right\|_H^2 \tag{10}$$

where C_s and C_t represent the self-representative coefficients of SD and TD, respectively, and l_s and l_t represent the sizes of the obtained SD and TD feature maps, respectively. This formula describes the distribution similarity of two domains by calculating the average difference of two domains in reproducing kernel Hilbert space (RKHS). The smaller the MMD value is, the more similar the two distributions are, and vice versa [21].

D. Auxiliary Loss

CSSRnet uses cross entropy loss to optimize the sorting task for marked samples on the branch of the SD. The contextual attention features that the SD learns through the self-presentation layer are classified through the fully connected layer with the Softmax function. The classification target is the common cross-entropy function mapping, and the predictive labels of the training samples are output. After that, the cross entropy formula is used to calculate the error between the predicted value and the real value

$$L_{\text{cls}} = -\frac{1}{n_s}\sum_{i=1}^{n_s}\sum_{c=1}^{S_c} y_i^c \log s_i^c \tag{11}$$

where y_i and s_i represent the label of SD and the prediction labels generated by SD branches, n_s represents the number of SD training samples, and S_c represents the number of ground object categories. This objective function can ensure that SD can learn to accurately distinguish the features of ground object types and then guide TD to better carry out the task of band selection.

On the TD, in the proposed framework CSSRnet, the contextual attention features generated by the TD dataset are also passed through the fully connected layer with softmax function to generate predictions for TD input samples. In order to control the contribution of target prediction to image classifier generalization, CSSRnet uses a low-density separation technique with entropy minimization (ENT) [39] to measure the class overlap of target samples

$$L_{\text{ent}} = -\frac{1}{n_t}\sum_{i=1}^{n_t} t_i \log t_i \tag{12}$$

where n_t is the number of training samples in the TD, and t_i represents the prediction labels generated for the branches of the TD.

By synthesizing the above loss function, the overall loss function of CSSRnet can be expressed as follows:

$$\mathcal{L} = L_{\text{da}} + L_{\text{MMD}} + L_{\text{cls}} + L_{\text{ent}}. \tag{13}$$

E. Optimal Bands Selection

The realization process of band self-representative can be seen as a band weighting process, in which the more important the band is, the greater the band's contribution to reconstruction. Therefore, CSSRnet selects the most representative combination of bands by ranking the self-representative coefficients of the bands. After the overall loss optimization is completed, each column of the self-representative coefficient is normalized first and then sorted by calculating the importance of each band. The bands with the highest importance rank are selected as the band selection result

$$\hat{C} = \frac{C_i}{\sum_{i=1}^{b} C_i} \tag{14}$$

$$b_i = \|\hat{C}_\iota\|_1 \tag{15}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:41:19 UTC from IEEE Xplore. Restrictions apply.

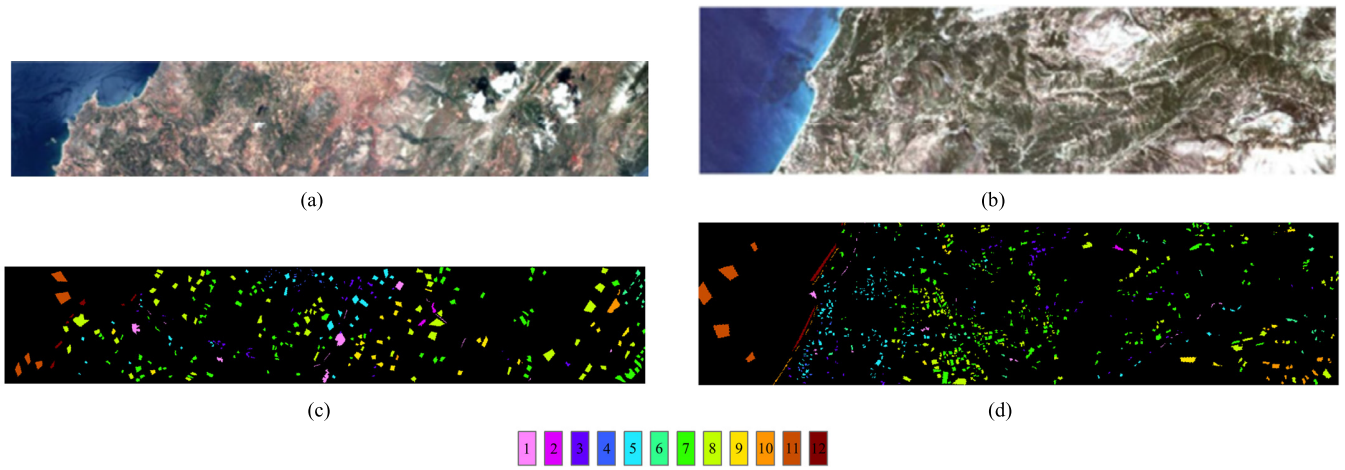


Fig. 3. False color image and ground truth of HyRANK dataset. (a) False color image of Dioni. (b) False color image of Loukia. (c) Ground truth of Dioni. (d) Ground truth of Loukia.

TABLE I
NUMBER OF SAMPLES IN HYRANK DATASET

Class		Number of samples	
No.	Name	Dioni(SD)	Loukia(TD)
1	Dense Urban Fabric	1262	206
2	Mineral Extraction Sites	204	54
3	Non Irrigated Arable Land	614	426
4	Fruit Trees	150	79
5	Olive Groves	1768	1107
6	Coniferous Forest	361	422
7	Dense Sderophyllous Vegetation	5035	2996
8	Sparce Sderophyllous Vegetation	6374	2361
9	Sparcely Vegetated Areas	1754	399
10	Rocks and Sand	492	453
11	Water	1612	1393
12	Coastal Water	398	421

TABLE II
NUMBER OF SAMPLES IN PAVIA DATASET

Class		Number of samples	
No.	Name	Pavia Center (SD)	Pavia University (TD)
1	Trees	7598	3064
2	Asphalt	9248	6631
3	Bricks	2685	3682
4	Bitumen	7287	1330
5	Shadows	2863	947
6	Meadows	3090	18649
7	Bare Soil	6584	5029

where b is the number of bands, and C_i is the ith column of self-representative coefficient C. Since the representative after $\hat{C}$ for normalized coefficient, b_i represents the importance of the ith band.

III. EXPERIMENTAL RESULTS

A. Experiment Datasets

To evaluate the effectiveness of the proposed method, experiments were conducted on three cross-scene hyperspectral datasets: HyRANK, Pavia, and ZY1-02D coastal wetland.

The HyRANK dataset was acquired by Hyperion sensors, which captured the images of the Earth's surface at 176 spectral bands. The HyRANK dataset was developed within the framework of the Scientific Initiatives of the International Society for Photogrammetry and Remote Sensing (ISPRS). This dataset comprises two scenes, namely, Dioni (SD) and Loukia (TD), with dimensions of 250 × 1376 and 249 × 945 pixels, respectively. The dataset encompasses a total of 12 common land classes. Table I provides information on the number of ground object samples contained in SD and TD. Fig. 3 displays both false color composite images and ground truth images for SD and TD.

The Pavia dataset was acquired by the Reflective Optics Spectrographic Image System (ROSIS) sensor over Pavia, comprising two landscape images: Pavia Center (SD) and Pavia University (TD). The former has a size of 1906 × 715 pixels with 102 spectral bands in the range of 430–860 nm. The latter contains 610 × 340 pixels with 103 spectral response bands. To maintain consistency with the number of bands in the Pavia Center, the last band of Pavia University was excluded [12]. The Pavia dataset has a spatial resolution of 1.3 m. Following the removal of two distinct types of ground objects, SD and TD maintain seven identical land cover classes. Table II outlines the number of ground objects present in both SD and TD. Fig. 4 displays the false color images of the Pavia dataset alongside its corresponding ground truth.

The ZY1-02D satellite, launched on September 12, 2019, represents China's first domestically developed commercial hyperspectral satellite [40]. The ZY1-02D coastal wetland dataset comprises two images, namely, Yellow River Delta and Yancheng, respectively. Yancheng Coastal Wetland is situated along the coast of the Yellow Sea in eastern Jiangsu, China. It boasts one of the world's largest and most comprehensive wetland ecosystems. The Yancheng data were acquired on September 6, 2020, with an image size of 1398 × 942 pixels.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:41:19 UTC from IEEE Xplore. Restrictions apply.

TABLE III
NUMBER OF SAMPLES IN ZY1-02D COASTAL WETLAND DATASET

Class		Number of samples	
No.	Name	Yellow River Delta (SD)	Yancheng (TD)
1	reed	310	244
2	spartina alterniflora	187	306
3	salt pan	306	901
4	suaeda	218	269
5	river	584	214
6	sea	4694	2076
7	idle land	459	650
8	building	398	451
9	bare land	87	235
10	paddy field	508	3026

After removing 19 bad bands, a total of 147 bands were available for subsequent applications. The Yellow River Delta, located at the estuary of the Yellow River in Dongying, Shandong, is the only one of the most complete, most typica,l and youngest wetland ecosystems in China. On June 28, 2020, Yellow River Delta was collected with a total of 119 bands retained after eliminating bad bands. The image size was measured as 1147 × 1600 pixels. In order to facilitate subsequent experiments, we further process the dataset by reducing the band number to 119 and retaining 10 types of ground objects. Fig. 5 shows the false-color composite map and the ground truth, while specific sample numbers are presented in Table III.

B. Experimental Setup

In this article, the proposed method is compared with six state-of-the-art methods, as well as using full bands. The six methods include two cross-scene band selection methods: CDIRF [30], CDWOASA [31], and four traditional methods: ISSC [41], E-FDPC [42], OCF [43], and BS-FC [32]. The parameter settings in the contrastive methods follow their defaults. Specifically, we set the number of iterations for CDIRF to 100, the kernel width to 0.5, and the stop criterion parameter to 0.00001. For CDWOASA, we set the number of iterations to 100 and the number of searchers to 5. In addition, a scalar parameter of 0.005 is assigned for ISSC. We designate top-rank cut and E-FDPC as the objective function and sorting method for OCF, respectively. The learning rate, regularization coefficient, and number of iterations for BS-FC are set to 0.01, 0.002, and 100, respectively. The learning rate of CSSRnet is set to 0.01 with a total of 200 iterations.

The cross-scene algorithms utilize SD labeled samples for model training. Contrarily, the remaining four traditional methods do not use labeled samples. In the HyRANK dataset, 150 labeled samples per class are randomly selected from SD for cross-scene algorithm training. For the Pavia dataset, the cross-scene methods are trained using 200 labeled samples per class in SD.

To evaluate the performances of the selected bands, we conduct classification experiments on TD using these bands. We choose one of the most commonly used support vector machine (SVM) as the classifier. The variance parameters and penalty factors in the SVM are determined based on threefold cross validation. We select 5% of labeled TD samples to train SVM and use the remaining of samples in TD for testing. In order to ensure fairness, each method is run ten times and the average accuracy of these runs is taken as the final result. Overall accuracy (OA) is selected to assess the classification accuracy of various methods. CSSRnet and BS-FC are implemented in Python 3.6, while other methods are implemented in MATLAB 2018a. The experiment is conducted on an Intel Core i7-10 700 2.90 GHz CPU with a RAM of 128 GB and NVIDIA GeForce RTX 3080.

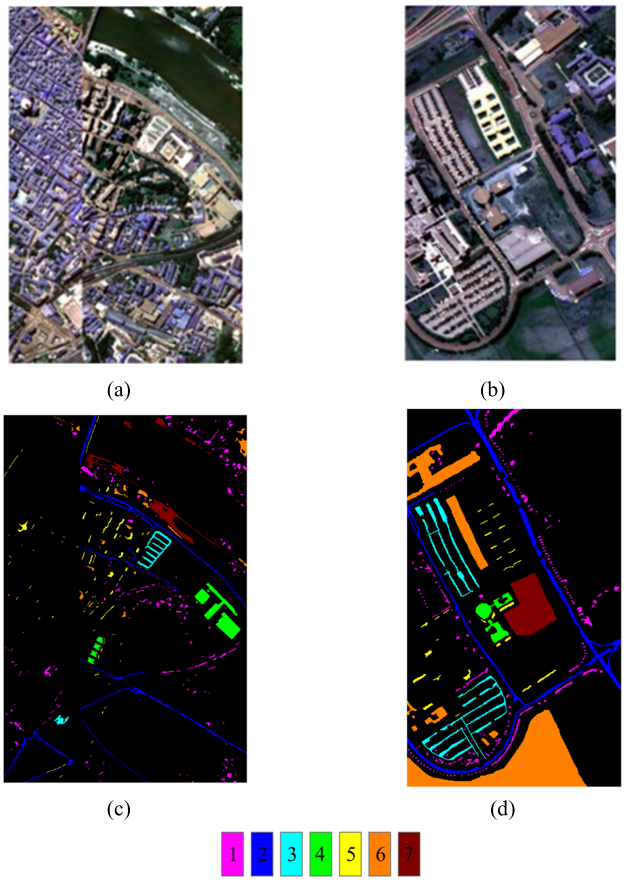


Fig. 4. False color image and ground truth of Pavia dataset. (a) False color image of Pavia Center. (b) False color image of Pavia University. (c) Ground truth of Pavia Center. (d) Ground truth of Pavia University.

C. Impact of Band Number on Classification Accuracy

To validate the band selection methods, we conducted experiments to assess the selected bands. Specifically, we analyzed the performance of different methods across varying numbers of selected bands. This section selects different combinations of bands with a step interval of 5 from 5 to 30 and conducts classification experiments to evaluate the efficacy of various methods. As AA, kappa, and OA exhibit similar results and curve fluctuations, only the OA classification result is presented. The classification accuracy result is illustrated in Figs. 6 and 7. To further analyze the impact of six methods, we selected 30 bands for subsequent experiments, and the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:41:19 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT GROUND OBJECTS IN HYRANK DATASET (%)

No	CSSRnet	ISSC	E-FDCP	OCF	BS-FC	CDIRF	CDWOASA
1	48.78±14.84	50.51±3.41	36.73±17.11	56.84±4.91	58.67±2.97	44.29±11.24	**60.31±5.22**
2	67.84±6.63	60.78±12.34	65.1±9.23	62.35±15.25	63.53±8.99	64.71±17.93	**71.76±11.8**
3	**80.44±3.11**	78.42±4.04	71.11±2.18	79.21±1.62	76±2.33	61.04±4.15	77.68±4.54
4	21.87±13.11	24.27±10.71	6.67±4.13	18.4±8.96	17.33±7.5	**25.07±12.99**	24.27±13.63
5	**87.26±2.55**	84.43±3.14	84.71±1.29	86.54±1.64	82±2.02	80.4±4.7	86.88±1.88
6	66.53±4.1	**66.88±6.04**	63.04±4.53	53.12±9.47	65.69±3.3	63.99±3.34	59.75±5.1
7	89.15±1.49	88.28±1.9	90.22±1.27	90.43±1.44	88.59±1.75	87.62±2.04	**90.48±1.96**
8	82.5±0.63	81.17±2.52	82.35±2.34	82.55±1.61	**83.87±1.24**	81.99±2.98	79.6±2.81
9	**60.74±8.83**	56.99±6.15	45.91±20.92	59.89±6.56	60.21±2.26	54.88±3.07	57.36±2.6
10	90.14±6.11	90.09±3.5	89.88±6.75	84.79±4.32	**90.98±3.19**	90.14±4.26	89.07±4.23
11	100±0	100±0	100±0	100±0	100±0	100±0	100±0
12	100±0	100±0	100±0	100±0	100±0	100±0	100±0
OA	**86.26±0.43**	84.33±0.39	83.95±1.01	85.1±0.51	84.85±0.7	83.24±0.43	85.05±0.27

TABLE V
CLASSIFICATION RESULTS OF DIFFERENT GROUND OBJECTS IN PAVIA DATASET (%)

No	CSSRnet	ISSC	E-FDCP	OCF	BS-FC	CDIRF	CDWOASA
1	91.32±0.84	**92.11±0.98**	89.82±0.86	90.92±0.99	91.08±1.64	88.7±1.18	90.28±1.56
2	**92.47±0.49**	91.4±0.73	91.32±0.87	91.52±0.36	92.07±0.59	88.16±1.64	90.6±1.19
3	95.56±1	96.47±0.66	94.95±1.39	**96.5±0.50**	96.2±1.39	95.6±0.58	95.1±1.13
4	78.88±3.1	79.56±3.5	**83.33±1.95**	81.03±2.36	78.5±1.34	64.8±7.99	82.01±2.43
5	**99.67±0.07**	99.2±0.38	99.36±0.28	99.29±0.3	99.24±0.37	98.87±0.96	99.47±0.28
6	97.95±0.36	**98.01±0.19**	96.85±0.67	97.36±0.34	97.69±0.24	96.37±0.24	96.55±0.7
7	**86.35±0.99**	84.53±1.35	82.78±1.3	84.56±1.39	83.32±1.76	79.29±1.56	75.51±3.69
OA	**94.18±0.28**	93.97±0.12	93±0.36	93.65±0.21	92.8±0.13	91.13±0.23	91.81±0.34

classification accuracy of various classes in both datasets is presented in Tables IV and V.

For the HyRANK dataset, as shown in Fig. 6, CSSRnet demonstrates significant advantages over the cross-scene band selection methods. It has consistently higher classification accuracy than CDIRF and CDWOASA. Notably, CDIRF exhibits significantly lower classification accuracy compared to both CSSRnet and CDWOASA. This indicates that CDIRF fails to effectively utilize the information from SD in assisting TD with band selection. While CSSRnet and CDWOASA exhibit a positive correlation between OA and number of bands, the OA curve for CDIRF remains relatively stable at a lower level. Compared to traditional methods, CSSRnet still holds advantages. Although OCF outperforms CSSRnet and achieves the highest classification accuracy when the number of bands is small (i.e., five versus ten bands), OCF's OA does not improve as quickly as that of CSSRnet. Among other traditional methods, E-FDPC's OA curve is similar to that of OCF, but its OA decreases when the number of bands exceeds 20. The performance of BS-FC and ISSC is suboptimal, while the OA exhibits significant fluctuations as the number of selected bands increases. From Table IV, it is evident that CSSRnet exhibits commendable performance in the classification of ground object types. It achieves the highest accuracy rate on six classes. This substantiates that CSSRnet can effectively identify diagnostic spectral bands to differentiate between various ground object types. CDWOASA achieves the optimal performance across five classes. Among traditional band selection methods, BS-FC demonstrates superior performance. The classification performance of all methods on the fourth class (fruit trees) in the HyRANK dataset is unsatisfactory, with an accuracy rate of only about 20%. This may be attributed to the limited number of samples available for this category in both source and TDs, resulting in misclassification. Meanwhile, the dataset contains a plethora of tree species with similar spectral curves, such as Class 4, Class 5, and Class 6, which poses a challenge to classification. CSSRnet ranks top in terms of accuracy among them as well. However, Class 11 and Class 12 have distinct spectral characteristics that are easily distinguishable from other ground objects. As a result, all methods achieved perfect classification accuracy.

In terms of the Pavia dataset, CSSRnet outperforms the other two cross-scene band selection methods. The classification accuracy of CSSRnet is consistently higher than that of the other two methods across all tested band combinations. This is similar to the performance in the HyRANK dataset. When the number of selected bands is limited, CDIRF shows a significant deviation from CDWOASA and CSSRnet. However, the OA curve slope of CDIRF is steeper, indicating that its classification accuracy improves more rapidly. In comparison to noncross-scene band selection methods, CSSRnet still retains certain advantages. The ISSC maintains the highest

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:41:19 UTC from IEEE Xplore. Restrictions apply.

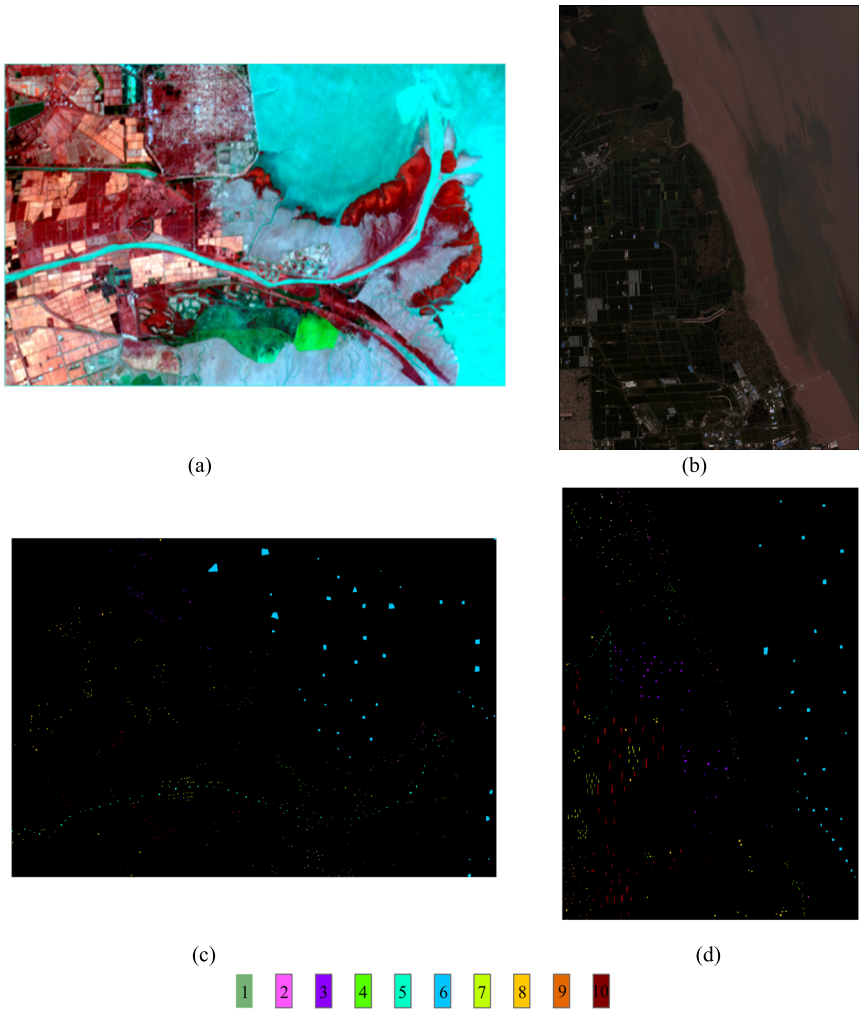


Fig. 5. False color image and ground truth of ZY1-02D coastal wetland dataset. (a) False color image of Dioni. (b) False color image of Loukia. (c) Ground truth of Dioni. (d) Ground truth of Loukia.

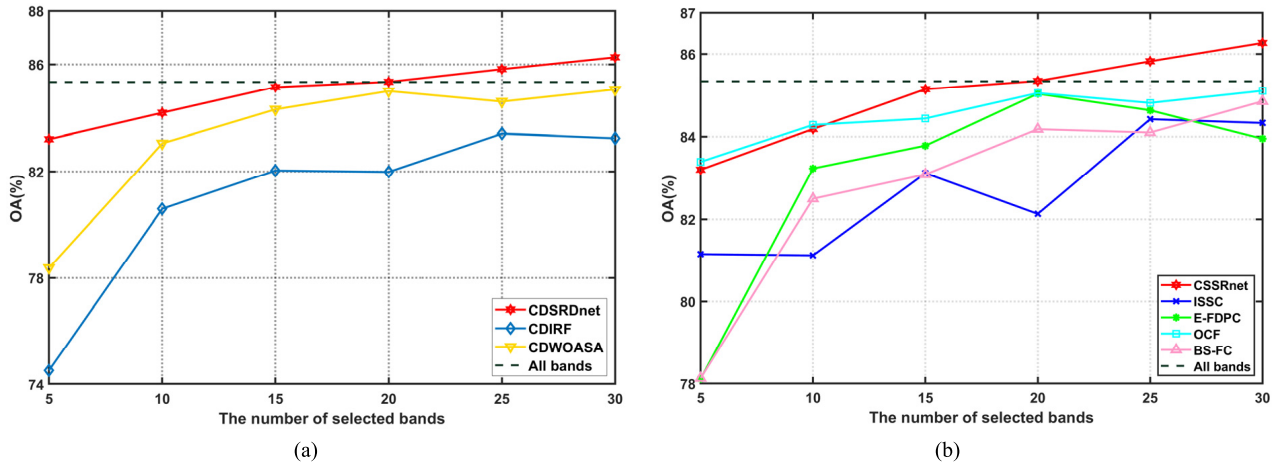
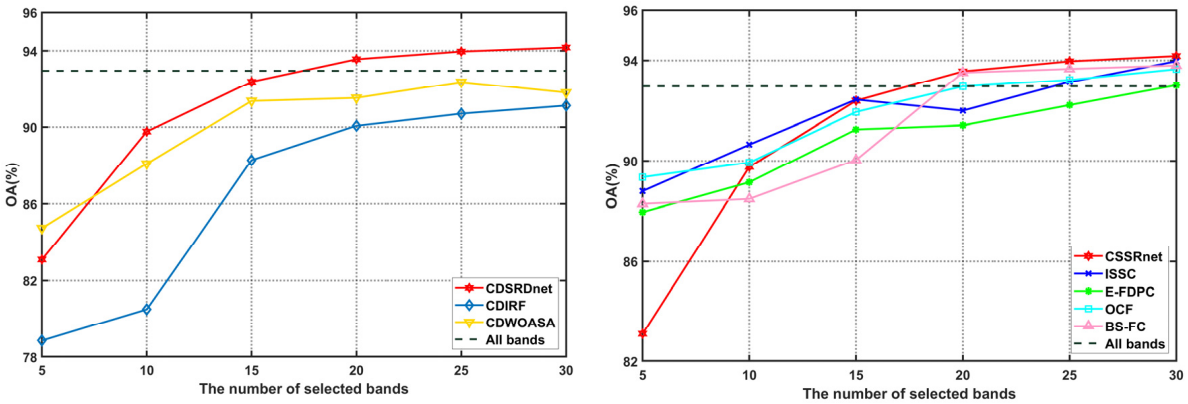


Fig. 6. Comparison of OA of different methods on HyRANK dataset with different number of bands. (a) Cross-scene methods. (b) Traditional methods.

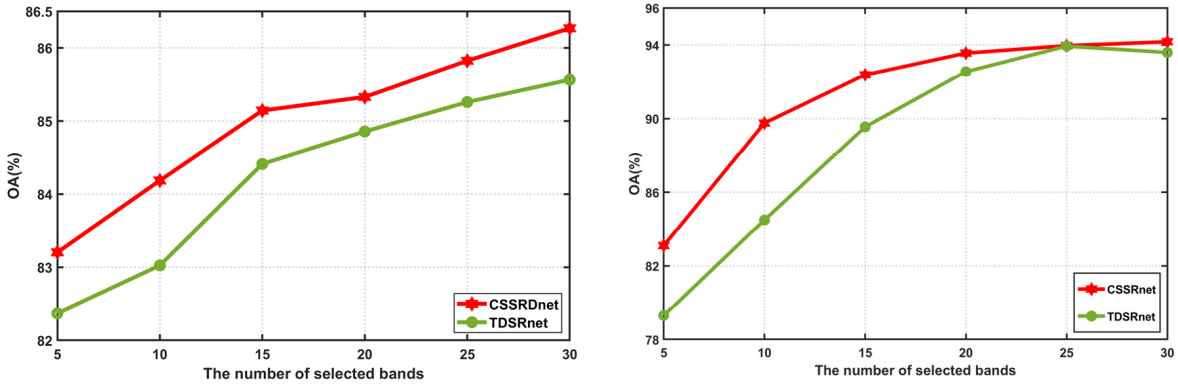
accuracy in ground object recognition and even achieves the best classification accuracy when there are ten bands. Although the classification performance of BS-FC is inferior when there are fewer than 15 bands, it shows outstanding classification results in scenarios with a large number of bands. OCF and E-FDPC demonstrate moderate performance, while the OA curve remains relatively stable. When the number of bands is 30, all methods outperform the classification performance of the full bands. However, with the exception of CSSRnet, these traditional methods did not exhibit superior performance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:41:19 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 7. Comparison of OA of different methods on Pavia dataset with different number of bands. (a) Cross-scene methods. (b) Traditional methods.



(a) (b)

Fig. 8. Influence of SD information. (a) HyRANK dataset. (b) Pavia dataset.

compared to the full band on two datasets, suggesting a certain gap in scene generalization ability among them. In the comparison of classification accuracy among different classes in the Pavia dataset, CSSRnet demonstrates superior performance by achieving the highest level of classification accuracy on three classes. ISSC achieves the highest classification accuracy for two classes. The classes in Pavia Center and Pavia University are distinction, with obvious spectral differences that allow for intuitive identification of diagnostic spectral bands. As a result, all seven methods demonstrated high classification accuracy for features in both locations.

D. Cross-Scene Validity Analysis

In order to verify the validity of SD information, an experiment is conducted on two datasets with and without such information. TDSRnet solely relies on TD information while disregarding SD data. The experiment is carried out under identical conditions as described above, and the results are presented in Fig. 8. From Fig. 5, it can be observed that as the feature dimension increases in the HyRANK dataset, the CSSRnet algorithm consistently outperforms TDSRnet in terms of OA. This provides evidence for the robust classification ability and stability of CSSRnet. Thus, the CSSRnet algorithm effectively improves the performance of TD band selection by utilizing the knowledge of SD in the HyRANK dataset to optimize the OA. For the Pavia dataset, TDSRnet algorithm's OA curve gradually converges to CSSRnet, despite CSSRnet still maintaining an advantage over TDSRnet as the number of selected bands continuously increases. This indicates that the CSSRnet algorithm proposed in this study effectively mitigates the impact of cross-scene variations, resulting in the selection of more robust features that are consistent across different scenes and ultimately improving classification accuracy for target scenes. Therefore, CSSRnet serves as an effective method for cross-scene band selection, which can significantly enhance the accuracy of classification across different scenes.

E. Cross-Scene Mapping of Coastal Wetlands

Coastal wetlands, as a crucial component of ecosystems, often exhibit intricate and diverse topography, making it challenging to access and obtain an adequate number of labeled samples. To validate the mapping efficacy of CSSRnet, we conducted an experiment using ZY1-02D satellite images acquired over China. Fig. 9 shows the mapping results of the Yancheng coastal wetland. CSSRnet applied to Yancheng data can effectively distinguish between reed and paddy fields, as demonstrated by the red ellipse, and exhibits superior results in the intersection zone of these two types of vegetation. OCF also performs well in this regard, accurately distinguishing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:41:19 UTC from IEEE Xplore. Restrictions apply.

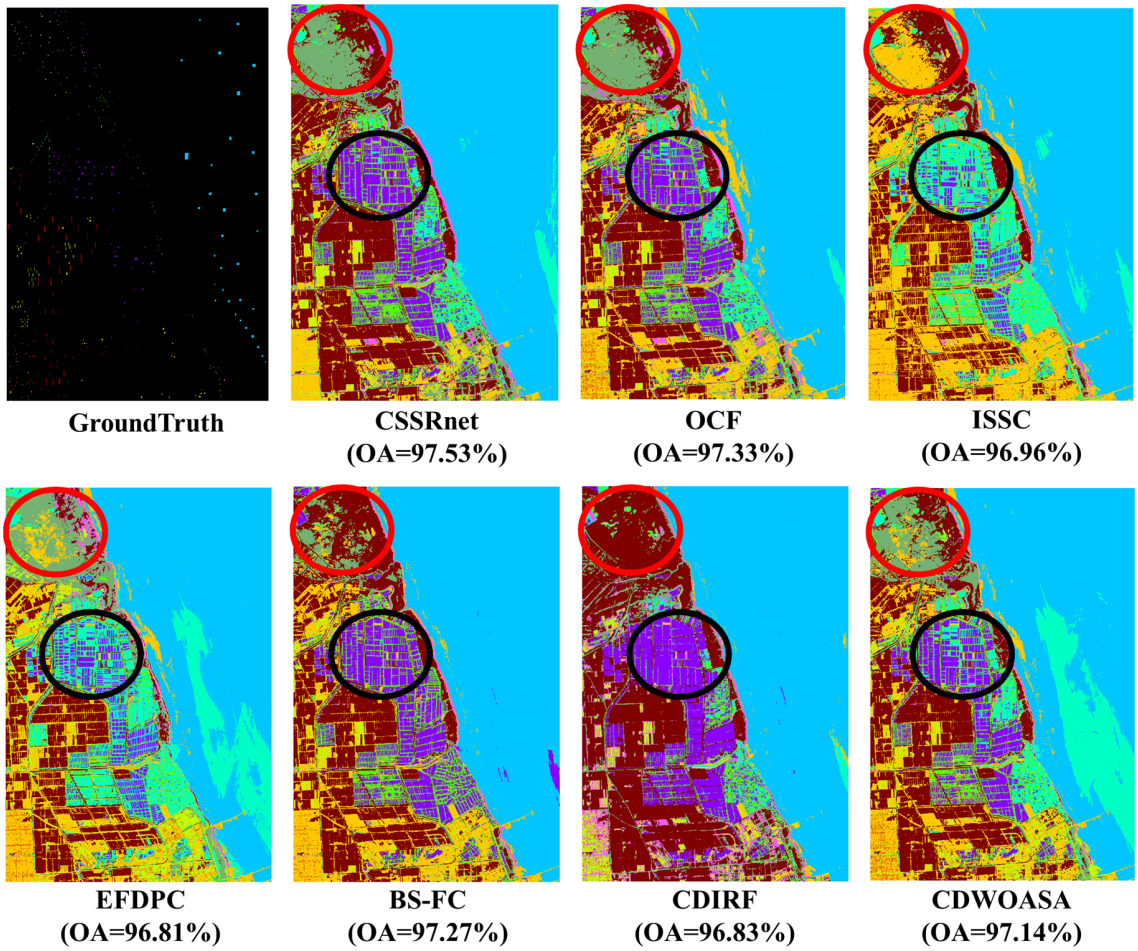


Fig. 9. Mapping results of different methods on Yancheng.

between reeds and paddy fields while clearly delineating their boundaries. However, most of the other comparison methods, such as BS-FC and CDIRF, show a significant degree of misclassification by erroneously categorizing reed into paddy fields with relatively similar spectral curves. A comparable result is observed in the black ellipse. In the black ellipse, OCF still performs well. However, it tends to misclassify more marine features as buildings in the land-sea junction. In contrast, CSSRnet is capable of maintaining a superior mapping effect. The proposed method incorporates a novel contextual attention mechanism that effectively integrates both long-range information and neighborhood information. This approach enables CSSRnet to accurately discriminate between ground objects and their boundaries. In contrast, other comparative methods either misclassify them as oceans or fail to capture the boundary information of salt pan. In summary, CSSRnet uses the rich sample knowledge of SD to facilitate TD's band selection and achieve exceptional coastal wetland mapping outcomes.

IV. Conclusion

This article proposes a CSSRnet for hyperspectral band selection. This method leverages substantial labels in the SD to steer band selection of TD. CSSRnet utilizes a deep learning framework to perform band selection for HSI. It incorporates self-representative layers with a novel contextual attention to capture potential features and self-representative coefficients of SD and TD. By utilizing the distribution alignment loss function and MMD, CSSRnet effectively mitigates data distribution discrepancies and feature disparities. Furthermore, by classification loss and ENT loss functions, CSSRnet enhances feature representation capabilities. Experimental results on hyperspectral datasets demonstrate that CSSRnet outperforms the comparison method in selecting a more representative band combination. It can even surpass the classification accuracy achieved by using all bands when the number of bands reaches a certain threshold. At the same time, CSSRnet has demonstrated superior mapping capabilities for coastal wetlands. In conclusion, it is an efficient band selection technique that exhibits excellent generalization properties in HSI. This approach can effectively boost the quality and efficiency of coastal wetland mapping while providing technical support for environmental monitoring and protection.

References

[1] J. W. P. Sun, H. Li, W. Li, X. Meng, C. Ge, and Q. Du, "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 10–43, Jun. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:41:19 UTC from IEEE Xplore. Restrictions apply.

[2] Q. Wang, Q. Li, and X. Li, "A fast neighborhood grouping method for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5028–5039, Jun. 2021.

[3] K. He et al., "A dual globallocal attention network for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens. Environ.*, vol. 60, pp. 1–13, 2022.

[4] R. J. Ellis and P. W. Scott, "Evaluation of hyperspectral remote sensing as a means of environmental monitoring in the St. Austell China Clay (Kaolin) region, Cornwall, U.K.," *Remote Sens. Environ.*, vol. 93, nos. 1–2, pp. 118–130, Oct. 2004.

[5] K. Liu et al., "Mapping coastal wetlands using transformer in transformer deep network on China ZY1–02D hyperspectral satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3891–3903, 2022.

[6] L. Zhizhong et al., "A review on the geological applications of hyperspectral remote sensing technology," in *Proc. 4th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Jun. 2012, pp. 1–4.

[7] S. Jia, S. Jiang, Z. Lin, N. Li, M. Xu, and S. Yu, "A survey: Deep learning for hyperspectral image classification with few labeled samples," *Neurocomputing*, vol. 448, pp. 179–204, Aug. 2021.

[8] B. Xu, X. Li, W. Hou, Y. Wang, and Y. Wei, "A similarity-based ranking method for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9585–9599, Nov. 2021, doi: 10.1109/TGRS.2020.3048138.

[9] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019, doi: 10.1109/mgrs.2019.2911100.

[10] W. Sun, J. Peng, G. Yang, and Q. Du, "Fast and latent low-rank subspace clustering for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3906–3915, Jun. 2020.

[11] W. Sun, J. Peng, G. Yang, and Q. Du, "Correntropy-based sparse spectral clustering for hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 3, pp. 484–488, Mar. 2020.

[12] X. Cao, T. Xiong, and L. Jiao, "Supervised band selection using local spatial information for hyperspectral image," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 3, pp. 329–333, Mar. 2016.

[13] W. Sun and Q. Du, "Graph-regularized fast and robust principal component analysis for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3185–3195, Jun. 2018.

[14] Y. Yuan, J. Lin, and Q. Wang, "Dual-clustering-based hyperspectral band selection by contextual analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1431–1445, Mar. 2016.

[15] W. Sun, L. Zhang, L. Zhang, and Y. M. Lai, "A dissimilarity-weighted sparse self-representation method for band selection in hyperspectral imagery classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 9, pp. 4374–4388, Sep. 2016.

[16] W. Sun, G. Yang, J. Peng, and Q. Du, "Hyperspectral band selection using weighted kernel regularization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3665–3676, Sep. 2019.

[17] J. Feng, L. Jiao, F. Liu, T. Sun, and X. Zhang, "Mutual-information-based semi-supervised hyperspectral band selection with high discrimination, high information, and low redundancy," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2956–2969, May 2015.

[18] J. Feng et al., "Dual-graph convolutional network based on band attention and sparse constraint for hyperspectral band selection," *Knowl.-Based Syst.*, vol. 231, Nov. 2021, Art. no. 107428.

[19] J. Feng et al., "Convolutional neural network based on bandwise-independent convolution and hard thresholding for hyperspectral band selection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4414–4428, Sep. 2021, doi: 10.1109/TCYB.2020.3000725.

[20] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. J. I. T. O. N. N. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, 2021.

[21] Y. Zhang, W. Li, R. Tao, J. Peng, Q. Du, and Z. Cai, "Cross-scene hyperspectral image classification with discriminative cooperative alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9646–9660, Nov. 2021.

[22] J. Geng, X. Ma, W. Jiang, X. Hu, D. Wang, and H. Wang, "Cross-scene hyperspectral image classification based on deep conditional distribution adaptation networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Mar. 2019, pp. 716–719.

[23] Y. Huang et al., "Two-branch attention adversarial domain adaptation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens. Environ.*, vol. 60, pp. 1–13, 2022.

[24] C. Zhong, J. Zhang, S. Wu, and Y. Zhang, "Cross-scene deep transfer learning with spectral feature adaptation for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2861–2873, 2020.

[25] Y. Ning, J. Peng, Q. Liu, Y. Huang, W. Sun, and Q. J. I. T. O. G. Du, "Contrastive learning based on category matching for domain adaptation in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens. Environ.*, vol. 61, pp. 1–14, 2023.

[26] Y. Yu, X. Xie, Y. Tang, and Y. Liu, "Feature selection for cross-scene hyperspectral image classification via improved ant colony optimization algorithm," *IEEE Access*, vol. 10, pp. 102992–103012, 2022.

[27] C. Persello and L. Bruzzone, "Kernel-based domain-invariant feature selection in hyperspectral images for transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2615–2626, May 2016.

[28] M. Ye, Y. Xu, H. Lu, K. Yan, and Y. Qian, "Cross-scene feature selection for hyperspectral images based on cross-domain information gain," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 4764–4767.

[29] M. Ye, Y. Xu, C. Ji, H. Chen, H. Lu, and Y. Qian, "Feature selection for cross-scene hyperspectral image classification using cross-domain ReliefF," *Int. J. Wavelets, Multiresolution Inf. Process.*, vol. 17, no. 5, 2019, Art. no. 1950039.

[30] C. Zhang, M. Ye, L. Lei, and Y. Qian, "Feature selection for cross-scene hyperspectral image classification using cross-domain I-ReliefF," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 5932–5949, 2021.

[31] J. Wang, M. Ye, F. Xiong, and Y. Qian, "Cross-scene hyperspectral feature selection via hybrid whale optimization algorithm with simulated annealing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2473–2483, 2021.

[32] Y. Cai, X. Liu, and Z. Cai, "BS-Nets: An end-to-end framework for band selection of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1969–1984, Mar. 2020, doi: 10.1109/TGRS.2019.2951433.

[33] B. Fang, Y. Li, H. Zhang, and J. Chan, "Hyperspectral images classification based on dense convolutional networks with spectral-wise attention mechanism," *Remote Sens.*, vol. 11, no. 2, p. 159, Jan. 2019, doi: 10.3390/rs11020159.

[34] Y. Li, T. Yao, Y. Pan, and T. Mei, "Contextual transformer networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 1489–1500, Feb. 2023.

[35] X. Wei, W. Zhu, B. Liao, and L. Cai, "Scalable one-pass self-representation learning for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4360–4374, Jul. 2019.

[36] Y. Cai, M. Zeng, Z. Cai, X. Liu, and Z. Zhang, "Graph regularized residual subspace clustering network for hyperspectral image clustering," *Inf. Sci.*, vol. 578, pp. 85–101, Nov. 2021.

[37] W. Sun, L. Tian, Y. Xu, D. Zhang, and Q. Du, "Fast and robust self-representation method for hyperspectral band selection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 11, pp. 5087–5098, Nov. 2017.

[38] M. Zeng, Y. Cai, Z. Cai, X. Liu, P. Hu, and J. Ku, "Unsupervised hyperspectral image band selection based on deep subspace clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 12, pp. 1889–1893, Dec. 2019.

[39] J. Wang, J. Chen, J. Lin, L. Sigal, and C. W. de Silva, "Discriminative feature alignment: Improving transferability of unsupervised domain adaptation by Gaussian-guided latent alignment," *Pattern Recognit.*, vol. 116, Aug. 2021, Art. no. 107943.

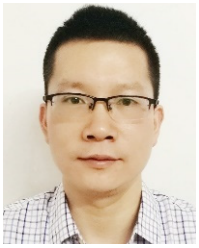
[40] W. Sun et al., "A simple and effective spectral–spatial method for mapping large-scale coastal wetlands using China ZY1–02D satellite hyperspectral images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 104, Dec. 2021, Art. no. 102572.

[41] W. Sun, L. Zhang, B. Du, W. Li, and Y. M. Lai, "Band selection using improved sparse subspace clustering for hyperspectral imagery classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2784–2797, Jun. 2015.

[42] S. Jia, G. Tang, J. Zhu, and Q. Li, "A novel ranking-based clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 88–102, Jan. 2016, doi: 10.1109/TGRS.2015.2450759.

[43] Q. Wang, F. Zhang, and X. Li, "Optimal clustering framework for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5910–5922, Oct. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:41:19 UTC from IEEE Xplore. Restrictions apply.

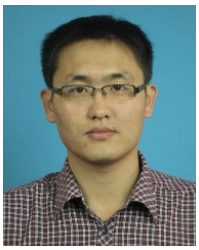


Weiwei Sun (Senior Member, IEEE) received the B.S. degree in surveying and mapping and the Ph.D. degree in cartography and geographic information engineering from Tongji University, Shanghai, China, in 2007 and 2013, respectively.

From 2011 to 2012, he was a Visiting Scholar at the Department of Applied Mathematics, University of Maryland College Park, College Park, MD, USA, with the famous Prof. John Benedetto to study the dimensionality reduction of Hyperspectral Images. From 2014 to 2016, he was a Post-Doctoral Researcher at the State Key Laboratory for Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, to study intelligent processing in hyperspectral imagery. From 2017 to 2018, he was a Visiting Scholar in hyperspectral image processing at the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. He is currently a Full Professor with Ningbo University, Ningbo, Zhejiang, China. He has published more than 80 journal articles. His research interests include hyperspectral image processing with machine learning.

Ke He received the B.S. degree in geographic information science from Shandong Jiaotong University, Jinan, China, in 2020, and the M.S. degree in geography from Ningbo University, Ningbo, China, in 2023.

His research interests include deep learning and the hyperspectral image process.



Gang Yang (Member, IEEE) received the M.S. degree in geographical information systems from the Hunan University of Science and Technology, Xiangtan, China, in 2012, and the Ph.D. degree from the School of Resource and Environmental Sciences, Wuhan University, Wuhan, China, in 2016.

He is currently an Associate Professor at Ningbo University, Ningbo, Zhejiang, China. His research interests include missing information reconstruction of remote sensing images, cloud removal of remote sensing images, and remote sensing time-series products temporal reconstruction.

Jiangtao Peng (Senior Member, IEEE) received the B.S. and M.S. degrees from Hubei University, Wuhan, China, in 2005 and 2008, respectively, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2011.

He is currently a Professor with the Faculty of Mathematics and Statistics, Hubei University. His research interests include machine learning and hyperspectral image processing.

Kai Ren (Student Member, IEEE) received the B.S. degree in humanistic geography and urban–rural planning from Shanxi Normal University, Xi'an, China, in 2018, and the M.S. degree in humanistic geography from Ningbo University, Ningbo, China, in 2021, where he is currently pursuing the Ph.D. degree with the Faculty of Electrical Engineering and Computer Science.

His research interests include remote sensing data processing and data analysis.

Jiancheng Li received the B.S., M.S., and Ph.D. degrees from the Wuhan Technical University of Surveying and Mapping, Wuhan, China, in 1987, 1990, and 1993, respectively.

From 1994 to 1995, he was a Visiting Scholar at the University of Texas Center for Space Research, Austin, TX, USA, and a Senior Visiting Scholar at Finland National Institute, Geodesy, Finland. From 2001 to 2001, he was a Senior Visiting Scholar at the German Research Center for Geosciences (GFZ), Potsdam, Germany, where he was a Researcher, from 2002 to 2003. From August 2008 to September 2008, he was a Senior Visiting Scholar at Germany Stuttgart University, Stuttgart, Germany. He is currently a Full Professor with the Faculty of Surveying and Mapping, Wuhan University, Wuhan. His research interests include geodesy and remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:41:19 UTC from IEEE Xplore. Restrictions apply.